Pregnancy Support System

Research Component: Virtual Nutritionist & GDM Action Engine

Project Proposal Report

Member 4 — Virtual Nutritionist Component
B.Sc. (Hons) Degree in Information Technology
Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Gestational Diabetes Mellitus (GDM) and nutritional deficiencies pose serious health risks to pregnant women and their unborn children in Sri Lanka. Existing digital nutrition tools fail to address these challenges adequately, relying on manual data logging that creates high user friction and offering static, one-size-fits-all dietary advice instead of personalised, context-aware intervention.

This research proposes a Multimodal Virtual Nutritionist a component of a four-member integrated maternal care ecosystem that serves as the system's Action Engine. The system eliminates tracking fatigue through three complementary logging modalities: Sinhala and English voice-to-text, computer vision food recognition using YOLOv8, and OCR-based nutrition label scanning for unknown packaged items. Before a mother eats, an AI risk model computes a pre-consumption Risk Score based on current blood glucose levels and the food's Glycemic Load, enabling predictive intervention rather than post-meal analysis. Context-aware hydration nudges fuse real-time weather temperature with dietary salt intake. Dynamic exercise recommendations adjust walking duration based on actual meal carbohydrate content, and live pedometer and accelerometer tracking confirms compliance. A gamified reward system sustains long-term engagement through health points and achievement badges.

The system integrates with three co-researchers whose components provide fetal growth data (Member 1 CTG Diagnosis), real-time biometric signals (Member 2 Risk Profiling), and mood analysis (Member 3 Mood Chatbot) to deliver holistic, personalised maternal care. Technical implementation employs Flutter, Python FastAPI, YOLOv8, Google Gemini API, and Firebase Firestore. A collaborative learning mechanism enables the system to learn new food products from one user and propagate that knowledge across the entire user base.

Expected outcomes include food recognition accuracy above 90%, demonstrably improved dietary logging compliance through gamification, and a measurable correlation between AI-guided exercise recommendations and stabilised post-meal blood glucose levels.

Keywords: Gestational Diabetes Management, Virtual Nutritionist, Multimodal Food Logging, Pre-Consumption Risk Assessment, Adaptive Exercise, Sri Lankan Maternal Health

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
BMI	Body Mass Index
CTG	Cardiotocography
GDM	Gestational Diabetes Mellitus
GI	Glycemic Index
GL	Glycemic Load
GPS	Global Positioning System
HR	Heart Rate
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
UX	User Experience

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1. INTRODUCTION

1.1 Background and Context

Pregnancy is one of the most nutritionally demanding periods in a woman's life. In Sri Lanka, a significant proportion of pregnant women suffer from iron-deficiency anaemia, insufficient calcium and folate intake, and poorly managed blood glucose all of which directly impact maternal and foetal outcomes [1]. Gestational Diabetes Mellitus (GDM) affects an estimated 14% of pregnancies globally and is rising in South Asian populations due to dietary patterns, sedentary lifestyles, and genetic predisposition [2].

Despite the proliferation of mobile health applications, existing digital nutrition tools fail pregnant women in three critical ways. First, they demand manual data entry for every meal, creating a phenomenon known as tracking fatigue where users progressively abandon consistent logging, rendering the data useless. Second, their dietary advice is static and generic rather than adapted to the mother's real-time physiological state. Third, virtually no tool is designed for the Sri Lankan context: local foods, Sinhala language support, and the specific dietary risks of the Sri Lankan diet are absent from existing solutions.

This research addresses these gaps through the development of a Multimodal Virtual Nutritionist the Action Engine of a four-member integrated maternal care research project. By fusing computer vision, voice recognition, real-time biometric data, environmental sensor inputs, and generative AI, the system transforms passive nutrition tracking into active, predictive, personalised intervention.

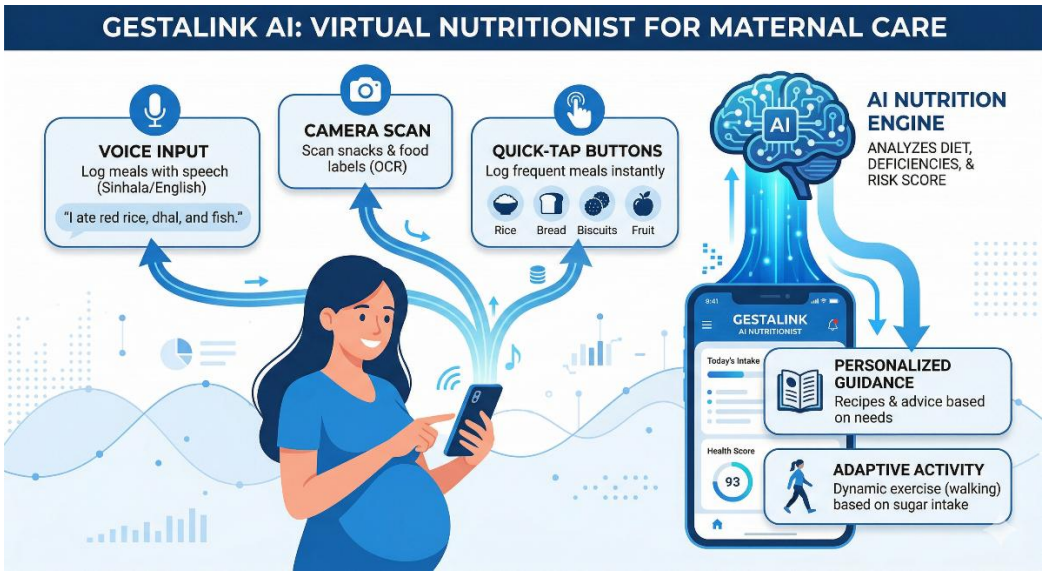


Figure 1: Multimodal Virtual Nutritionist — System Overview

1.2 Literature Review

1.2.1 Gestational Diabetes and Digital Health Interventions

Retnakaran et al. (2019) demonstrated that early dietary intervention in GDM patients significantly reduces the risk of macrosomia and neonatal hypoglycaemia [3]. However, their study relied on paper-based food diaries, a method associated with over 40% under-reporting of caloric intake. Bhavadharini et al. (2020) conducted a large-scale study of GDM prevalence in South Asian populations, confirming that post-meal blood glucose management is the most clinically significant variable yet most apps only prompt users after meals rather than before [4].

Mackillop et al. (2018) trialled a digital tool for blood glucose monitoring in gestational diabetes patients and found that real-time feedback improved glycaemic control compared to standard care, but noted that compliance dropped sharply when data entry required more than 30 seconds per interaction [5]. This directly motivates our multimodal logging approach.

1.2.2 Computer Vision for Food Recognition

Meyers et al. (2015) introduced the Im2Calories system, using deep convolutional neural networks to estimate caloric content from meal photographs with 75% accuracy on a controlled dataset. Their work established food photo analysis as a viable nutrition tracking modality [6]. More recently, YOLOv8 the current state of the art in real-time object detection has demonstrated accuracy above 91% on food item classification in restaurant and packaged food contexts [7].

Ciocca et al. (2017) applied CNNs to a dataset of 1,500 dishes and achieved 84% recognition accuracy, but noted significant accuracy degradation on visually similar items — a limitation our OCR-based nutrition label fallback directly addresses [8].

1.2.3 Glycemic Index, Glycemic Load, and Pre-Meal Risk Assessment

Jenkins et al. (2002) formally established the Glycemic Load (GL) concept as a more clinically meaningful metric than Glycemic Index alone, as it accounts for both food quality and serving size [9]. Wolever (2013) demonstrated that GL-based dietary interventions produced superior blood glucose stability in Type 2 and gestational diabetic patients compared to carbohydrate counting alone [10].

Critically, no existing consumer application has implemented pre-consumption risk scoring where the system evaluates a food's suitability before the user eats, factoring in their current glucose levels. This represents the primary research novelty of this component.

1.2.4 Gamification in Health Applications

Hamari et al. (2014) conducted a systematic review of 24 peer-reviewed studies on gamification in health contexts and found that point systems, achievement badges, and

progress visualisation increased engagement metrics by an average of 30% compared to non-gamified equivalents, provided the motivational design aligned with Self-Determination Theory principles of autonomy, competence, and relatedness [11].

1.2.5 Comparison of Existing Classification Approaches

Study	Method	Accuracy	Key Limitation
MyFitnessPal (2022)	Manual Logging + Barcode	N/A	No GDM logic; no local foods
Meyers et al. (2015)	CNN Food Photo Analysis	75%	No nutritional calculation; generic foods
Mackillop et al. (2018)	Digital Blood Glucose Tracker	N/A	No dietary advice integration
YOLOv8 (Redmon, 2023)	Real-Time Object Detection	91%+	Not adapted for Sri Lankan dishes
Wolever (2013)	GL-Based Dietary Advice	N/A	Manual process; not real-time
Hamari et al. (2014)	Gamified Health App	30% lift	SDT principles often ignored in design

1.3 Research Gap

Based on the literature review and analysis of existing systems, four critical gaps emerge:

#	Gap	Why It Matters	Our Solution
1	**No pre-consumption risk scoring exists**	Existing apps analyse meals after eating — too late to prevent a dangerous glucose spike	AI Risk Score computed before eating, fusing current glucose and food Glycemic Load
2	**No contextual data fusion for nutrition**	Static advice ignores real-time physiology, weather, and environmental conditions	Links weather temperature + salt intake for hydration; blood glucose for exercise adaptation
3	**High tracking fatigue in existing apps**	40%+ drop-off in manual logging within 2 weeks reduces clinical utility to zero	Trimodal logging (voice + vision + OCR) with Quick Log to reduce interaction to 1 tap

#	Gap	Why It Matters	Our Solution
4	**No Sri Lankan dietary context in any tool**	Foreign food databases are clinically useless for local GDM management	Sri Lankan food database with Sinhala voice support and collaborative learning from users

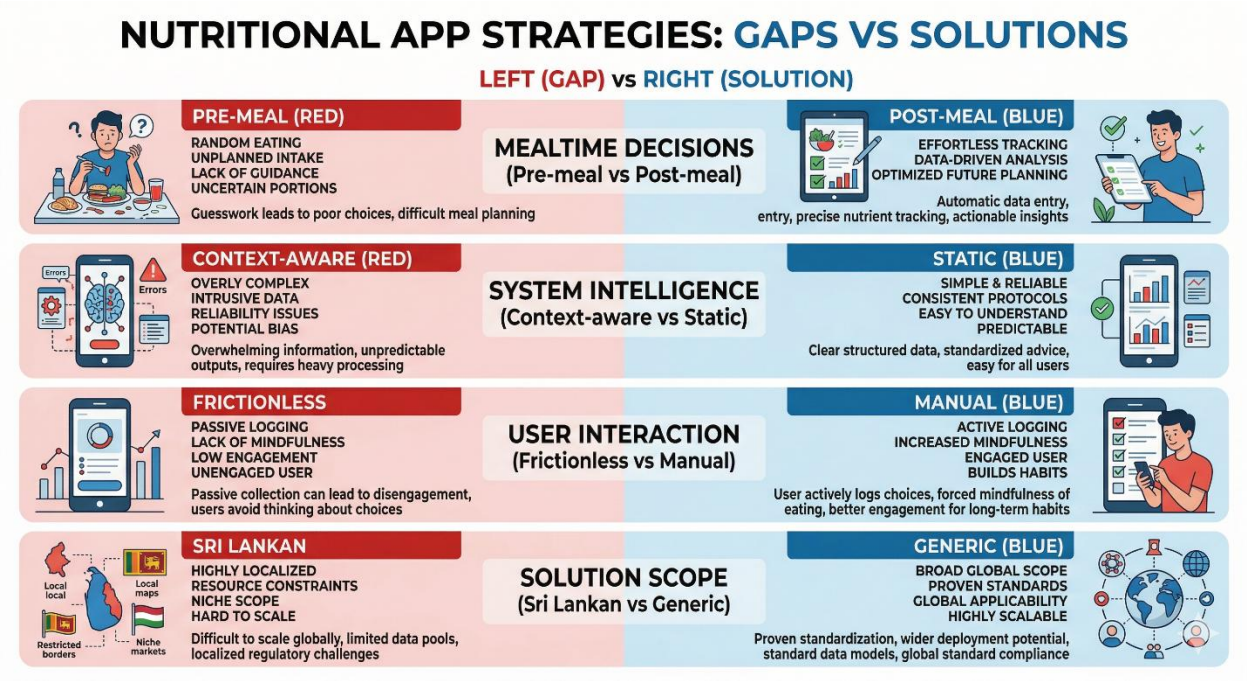


Figure 2: Research Gap Summary Diagram

2. OBJECTIVES

2.1 Main Objective

Design and implement a Multimodal Virtual Nutritionist that eliminates manual dietary tracking friction through computer vision, voice input, and OCR; delivers pre-consumption AI risk scoring for gestational diabetes management; adapts exercise recommendations dynamically to meal carbohydrate content; and integrates with fetal growth, biometric risk, and mood data from co-researchers achieving food recognition accuracy above 90% and measurably improved dietary compliance through gamification.

2.2 Specific Objectives

- **SO1 — Multimodal Food Logging:** Implement three complementary data acquisition tracks (voice, computer vision, OCR) to enable frictionless dietary tracking, targeting a system usability score above 80 on the SUS scale.
- **SO2 — Pre-Consumption Risk Scoring:** Develop an AI risk model that computes a personalised Risk Score before the mother eats, using current blood glucose and Glycemic Load, validated against clinical GDM dietary guidelines.
- **SO3 — Nutritional Deficiency Detection:** Analyse micro-nutrient intake (Iron, Calcium, Folate) to identify deficiencies before clinical presentation and generate personalised recipe recommendations to correct them.
- **SO4 — Adaptive Exercise Intervention:** Dynamically compute required walking duration based on meal carbohydrate content and verify completion using smartphone accelerometer and pedometer data.
- **SO5 — Contextual Hydration Logic:** Fuse real-time weather temperature data and dietary salt intake to generate context-aware hydration nudges superior to simple timed reminders.
- **SO6 — Gamification & Compliance:** Implement a health points and achievement badge system designed around Self-Determination Theory principles, demonstrating statistically significant improvement in logging compliance compared to a non-gamified baseline.
- **SO7 — Collaborative Learning Database:** Build a mechanism by which newly scanned food packages are learned from one user's OCR scan and added to the shared Sri Lankan food database for all users.

3. METHODOLOGY

3.1 Research Approach

This research follows the Design Science Research (DSR) methodology, appropriate for projects that produce a functional technological artefact and evaluate it through rigorous structured testing and user trials:

- Phase 1 — Problem Identification: Analyse existing maternal nutrition apps, clinical GDM literature, and user friction patterns. Identify the Sri Lankan dietary context gap.
- Phase 2 — Design: Architect the trimodal logging system, AI risk engine, adaptive exercise logic, and gamification framework. Define integration contracts with co-researchers.
- Phase 3 — Implementation: Build and train the YOLOv8 food recognition model, develop the risk scoring algorithm, implement the Flutter frontend, and connect the FastAPI backend.
- Phase 4 — Evaluation: Conduct technical accuracy testing, user compliance trials with and without gamification, and clinical correlation analysis between AI exercise guidance and blood glucose stabilisation.

3.2 Multimodal Dietary Data Acquisition

To maximise dietary logging compliance, the system provides three distinct input tracks, each designed for a specific user context:

Track	Method	Key Benefit
Main Meal Logging	Sinhala/English voice-to-text + manual text entry. Caregivers may log on behalf of the mother with explicit permission.	Natural interaction; no typing required; family involvement reduces burden on mother.
Snack Recognition	YOLOv8 computer vision identifies snack items from camera. Unknown items trigger OCR scan of nutrition label to extract sugar and calorie data.	Eliminates lookup time; learns new products for all users.
Quick Log	'Recent Items' and 'As Usual' single-tap buttons for frequently consumed foods. Voice shortcut 'Same as yesterday breakfast' supported.	Reduces interaction from 60+ seconds to under 5 seconds.

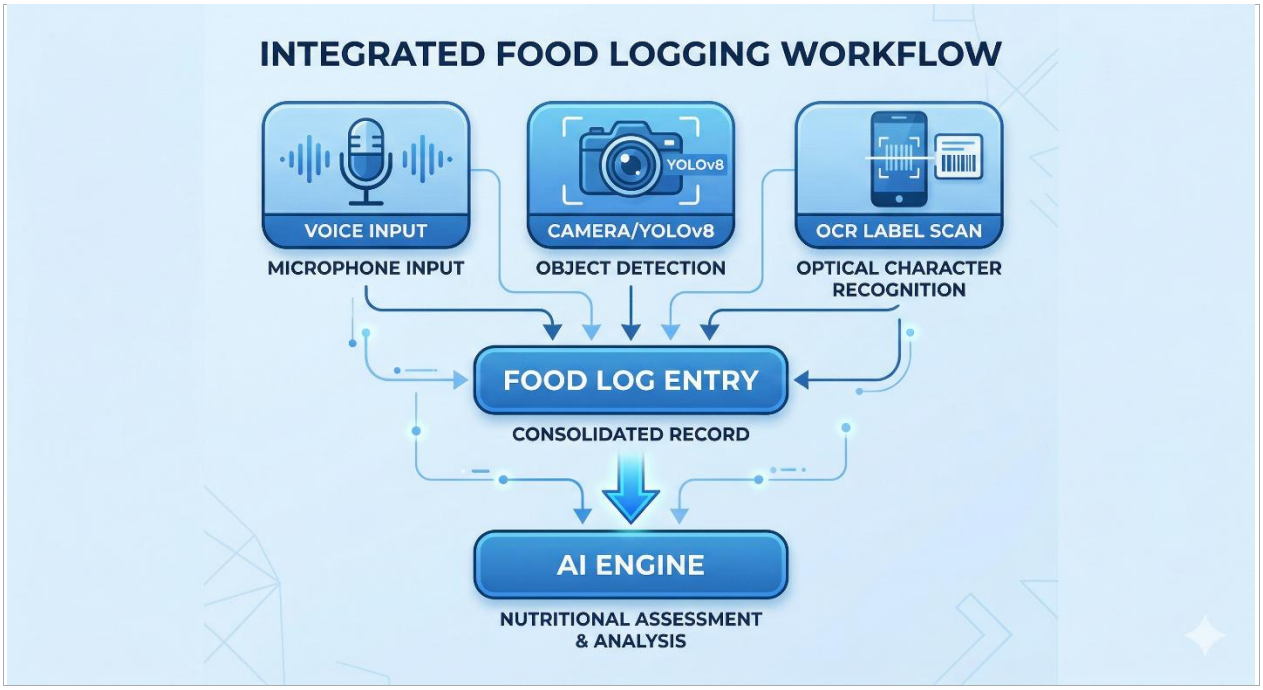


Figure 3: Trimodal Dietary Logging — Data Flow

3.3 The AI Nutrition Engine

3.3.1 Nutritional Deficiency Detection

The system maintains a rolling 7-day nutritional profile for each user, tracking key micro-nutrients critical in pregnancy: Iron (target: 27mg/day), Calcium (1000mg/day), and Folate (600mcg/day). When cumulative intake falls below 80% of the recommended daily allowance for two or more consecutive days, the system flags a potential deficiency and triggers a recipe recommendation from the Virtual Chef module tailored to address that specific deficiency.

3.3.2 Pre-Consumption Risk Scoring

The central research novelty is the shift from post-meal analysis to pre-consumption risk assessment. Before eating, the mother queries 'Can I eat this?'. The AI computes a Risk Score using the following formula:

$$\text{Risk Score} = (\text{Glycemic Load} \times \text{Current Blood Glucose}) \div \text{Insulin Sensitivity Factor}$$

Risk scores are categorised into three tiers Safe (< 4.0), Caution (4.0–7.0), and High Risk (> 7.0) with colour-coded visual feedback. When a High Risk score is produced, the system does not refuse the food but instead recommends a portion reduction or a safer alternative from the Sri Lankan food database that achieves a similar satiety level with a lower Glycemic Load.

Critically, the Insulin Sensitivity Factor is not static. It is personalised using the mother's recent blood glucose response history (provided via the Member 2 Risk Profiling integration) and is updated weekly using a simple Bayesian estimation approach. This ensures the risk score remains clinically meaningful as the mother's glucose sensitivity changes throughout pregnancy.

3.3.3 Context-Aware Hydration Logic

Rather than simple timed reminders (e.g., 'drink water every 2 hours'), the system fuses two independent signals to generate situation-appropriate hydration nudges:

- Ambient temperature via OpenWeatherMap API: When local temperature exceeds 30°C, hydration interval shortens by 25%.
- Dietary sodium intake: When cumulative sodium from logged meals exceeds 1500mg, an urgent hydration alert is triggered regardless of time since last reminder.

This contextual approach is particularly important for Sri Lankan climate conditions, where high humidity and heat combined with sodium-rich traditional dishes create compounding dehydration risk during pregnancy.

3.3.4 Virtual Chef — Personalised Recipe Recommendations

When a nutritional deficiency or an unsafe food choice is detected, the Virtual Chef module powered by Google Gemini API generates a personalised recipe recommendation. The generation prompt is constrained by three parameters: the detected deficiency or GDM constraint, the ingredients the mother reports as available in her kitchen, and her taste preferences established during onboarding. Recipes are provided in both Sinhala and English.

3.4 Adaptive Behavioural Intervention — Exercise Logic

Post-meal walking is a clinically validated intervention for reducing post-prandial blood glucose spikes in GDM patients [12]. Rather than generic recommendations, the system calculates a dynamic walking target after each meal:

```
Walking Duration (min) = (Net Carbohydrates consumed × 0.5) +
                         Base Duration (10 min)
```

The smartphone's built-in accelerometer and pedometer are used to track whether the mother completes the recommended 'Power Walk'. Step count, walking pace, and duration are logged. If the target is not met within 90 minutes of the meal, the system sends a gentle reminder framed as a gamification quest ('Complete your walk to earn 50 Health Points!') rather than a clinical warning.

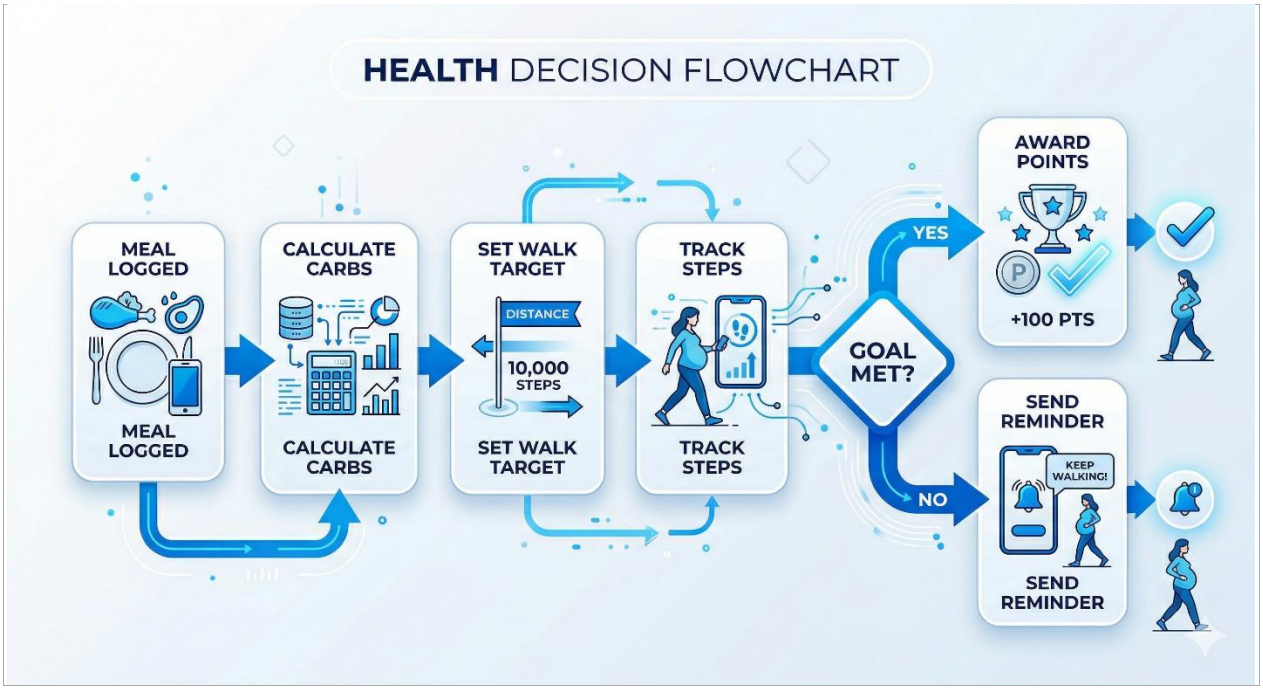


Figure 4: Adaptive Exercise Logic — Decision Flow

3.5 Collaborative Ecosystem Integration

This component does not operate in isolation. It serves as the Action Engine of a four-researcher ecosystem:

Researcher	Component	Integration with Virtual Nutritionist
Member 1	CTG Diagnosis	Foetal growth data (weight estimate, gestational age) informs calorie and protein targets. Mothers carrying larger foetuses receive higher protein recommendations.
Member 2	Risk Profiling	Real-time Blood Glucose, Heart Rate, and Blood Pressure from wearables trigger the Exercise module and Emergency Advice when readings fall outside safe ranges.
Member 3	Mood Chatbot	Detected emotional state adjusts the tone of nutritional advice delivery — soothing language for anxious moods; motivating language for positive states.
Member 4 (This)	Virtual Nutritionist	Dietary data, exercise completion, and nutritional status feed back to Member 2 for holistic risk profile updates.

3.6 Data Collection and Ethical Considerations

Training data for the YOLOv8 food recognition model will be sourced from: (1) the Food-101 dataset supplemented with 500+ manually photographed Sri Lankan dishes; (2) nutrition label photographs collected with user consent during beta testing; and (3) synthetic augmentation (rotation, brightness variation, partial occlusion) to improve model robustness.

All personal health data blood glucose readings, dietary logs, GPS data for weather API calls is stored with user-specific encryption keys in Firebase Firestore. Data is never shared with third parties and users retain full deletion rights. Research will be conducted in accordance with SLIIT ethical guidelines, and any clinical trial involving pregnant women will receive prior institutional ethics approval.

3.7 Validation Strategy

Test	Method	Success Criteria
Food Recognition Accuracy	Confusion Matrix on held-out test set of 500 images (Sri Lankan and international foods)	Overall accuracy ≥ 90%; F1-score ≥ 0.88
OCR Label Accuracy	100 pre-photographed nutrition labels scanned; values compared to ground truth	Nutrient extraction accuracy ≥ 95%
Risk Score Clinical Validity	50 risk scores compared against dietitian assessment of the same meal/glucose combinations	Agreement ≥ 85% on Safe/Caution/High Risk triage
Logging Compliance	2-week comparison: gamified group vs control group (n=20 each)	Gamified group logs ≥ 30% more meals per week
Exercise Adherence	Pedometer data vs recommended walk duration for 30 participants over 1 week	≥ 70% of recommended walks completed when gamification active
Blood Glucose Correlation	Pre and post-walk glucose readings for 20 participants over 4 weeks	Statistically significant reduction in post-prandial glucose spike ($p < 0.05$)
Response Time	API load test with 100 concurrent requests	95th percentile response ≤ 2 seconds

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The Virtual Nutritionist component consists of four interconnected architectural layers, each handling a distinct concern:

Layer	Component	Responsibility
1 — Input	**Multimodal Acquisition Layer**	Voice STT, YOLOv8 camera recognition, OCR label scanning, Quick Log. All paths converge to a unified Food Log Entry object.
2 — Intelligence	**AI Nutrition Engine (FastAPI + TensorFlow)**	Deficiency detection, Risk Score computation, hydration nudge logic, Virtual Chef recipe generation via Gemini API.
3 — Action	**Behavioural Intervention Layer**	Dynamic exercise target calculation, pedometer/accelerometer tracking, gamification reward processing, accountability alerts.
4 — Sync	**Data & Integration Layer (Firebase Firestore)**	Real-time sync between mother, family caregiver, doctor dashboard, and co-researcher data streams.

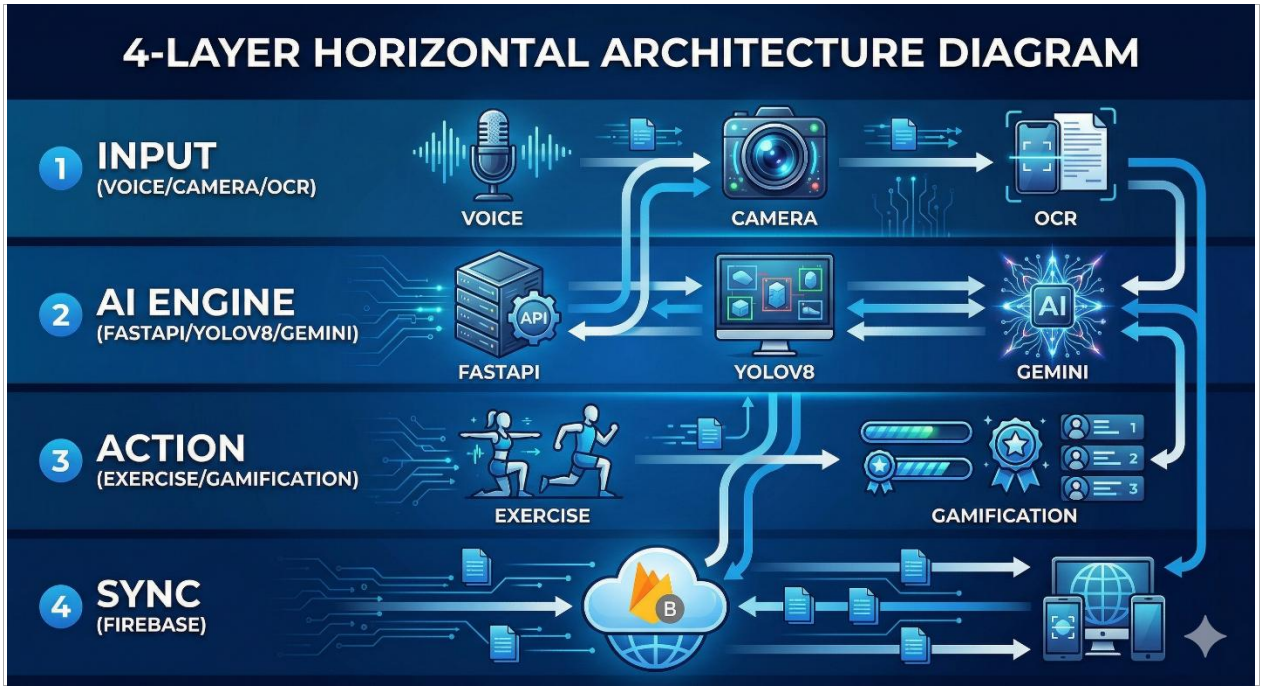


Figure 5: High-Level System Architecture — Four Layers

4.1 Technical Stack

Layer	Technology	Reason for Selection
Frontend	Flutter (Dart)	High-performance sensor access (GPS, pedometer, accelerometer); single codebase for Android and iOS.
Backend	Python 3.11 + FastAPI	Essential for running AI/ML models (TensorFlow/PyTorch); async support for real-time data processing.
AI — Vision	YOLOv8 (Ultralytics)	State-of-the-art real-time object detection; lightweight enough for on-device inference.
AI — Language	Google Gemini API	Generative recipe creation and conversational nutritional advice with Sinhala language support.
Database	Firebase Firestore	Real-time bidirectional sync between mother, family, doctor, and co-researcher data streams.
Environmental	OpenWeatherMap API	Real-time local temperature for context-aware hydration logic.
Speech	Google STT / flutter_speech	Sinhala and English voice-to-text for frictionless meal logging.

4.2 Scalability and Security

All health data is encrypted at rest using AES-256 and in transit using TLS 1.3. Firebase security rules enforce that each user can only read their own health records. The YOLOv8 model runs on-device for offline scenarios, falling back to server-side inference for higher accuracy when connectivity is available. The collaborative learning pipeline for new food products includes a human-in-the-loop review step before a community-submitted food entry is published to the shared database, preventing deliberate or accidental data poisoning.

5. GAMIFICATION & ENGAGEMENT DESIGN

The gamification system is designed according to Self-Determination Theory (SDT) [11], targeting the three fundamental psychological needs that drive sustained intrinsic motivation:

SDT Need	Game Mechanic	Implementation
Autonomy	Flexible Quest System	Mothers choose their daily health quests from a personalised menu rather than receiving fixed tasks.
Competence	Health Points + Level Progression	Points earned for logging, completing walks, hitting hydration targets. 10 levels from 'Wellness Starter' to 'Golden Mom'.
Relatedness	Family & Community	Family caregivers earn 'Support Points' for logging on the mother's behalf. Weekly leaderboard among consenting friends/family.

Accountability alerts serve a dual function. When dietary logging stops for more than 48 hours, the AI warns the mother that prediction accuracy is degrading due to insufficient data. The doctor's dashboard simultaneously receives a low-data flag, ensuring clinical oversight is maintained even when the mother is disengaged. Critically, these alerts are framed as helpful system limitations rather than user failures, to avoid shame-based disengagement.

6. RESEARCH NOVELTY

This research contributes four distinct novelties not found in any existing maternal nutrition application:

#	Novelty	Description
1	**Pre-Consumption Risk Assessment**	No existing app computes a risk score before the user eats. This shifts intervention from reactive (post-meal tracking) to predictive (pre-meal decision support) — a fundamentally different clinical model.
2	**Contextual Logic Fusion**	Linking external environmental data (weather temperature) with internal physiological data (blood glucose, sodium intake) to trigger dietary and hydration actions is unprecedented in maternal nutrition apps.
3	**Collaborative Food Database Learning**	A community-driven mechanism where one user's OCR scan of an unknown food packet enriches the database for all users, solving the Sri Lankan food coverage problem at scale without central data entry.
4	**Ecosystem Integration**	Unlike standalone apps, this component is architecturally designed to receive and emit data across a four-researcher maternal care ecosystem, enabling holistic interventions that account for foetal growth, biometric risk, and emotional state simultaneously.

7. USER REQUIREMENTS

7.1 Stakeholder Identification

Stakeholder	Interaction	Primary Need
Pregnant Mothers (Primary)	Directly uses app for logging, risk queries, and exercise guidance	Minimal effort logging; clear, timely dietary guidance; safe blood glucose management
Family Caregivers	Logs food on mother's behalf; receives compliance alerts	Simple interface; confidence that advice is clinically grounded
Gynaecologists / Dietitians	Views aggregated nutritional data and compliance reports on doctor dashboard	Reliable, complete data; alerts when clinical thresholds are approached
Co-Researcher Systems	Provides biometric data (M2), foetal data (M1), mood data (M3) via API	Stable, documented API contracts; real-time data availability

7.2 Functional Requirements

ID	Requirement	Priority
FR-01	Log meals via Sinhala/English voice-to-text with automatic food identification	**Critical**
FR-02	Identify food items from camera photographs using YOLOv8	**Critical**
FR-03	Extract nutritional data from food package labels via OCR and add to shared database	**Critical**
FR-04	Compute pre-consumption Risk Score using current blood glucose and food Glycemic Load	**Critical**
FR-05	Detect micro-nutrient deficiencies and trigger recipe recommendations	**Critical**
FR-06	Generate context-aware hydration nudges based on temperature and sodium intake	High
FR-07	Calculate post-meal walking targets based on meal carbohydrate content	**Critical**

ID	Requirement	Priority
FR-08	Track exercise completion using pedometer and accelerometer	High
FR-09	Award Health Points and Badges for logging compliance and exercise completion	High
FR-10	Sync dietary and exercise data in real-time with doctor dashboard and co-researcher APIs	**Critical**
FR-11	Allow family caregivers to log food data on behalf of the mother	Medium
FR-12	Send accountability alerts to mother and doctor when logging gaps exceed 48 hours	High

7.3 Non-Functional Requirements

Category	Requirement
Accuracy	Food recognition ≥ 90%; OCR nutrient extraction ≥ 95%; Risk Score clinical agreement ≥ 85%
Response Time	Risk Score result displayed within 3 seconds; voice-to-text transcription within 2 seconds
Offline Capability	YOLOv8 food recognition and Quick Log function without internet; data syncs when connectivity restores
Security	AES-256 data encryption at rest; TLS 1.3 in transit; user-scoped Firebase security rules
Usability	System Usability Scale (SUS) score ≥ 80; Sinhala and English UI supported
Availability	Application available 99.5% of the time; Firebase provides automatic failover
Privacy	No health data shared with third parties; users retain full deletion rights
Explainability	Risk Score display shows contributing factors (current glucose, food GL, sensitivity) transparently

8. COMMERCIALIZATION PLAN

8.1 Target Market

The primary market is healthcare providers government maternity hospitals, private obstetric clinics, and nutritional support programmes targeting GDM patients. The secondary market is direct-to-consumer through app stores, targeting pregnant women who self-manage their nutrition. The tertiary market is health insurance companies who benefit from preventable complication reduction.

8.2 Value Proposition

The system moves maternal nutritional care from episodic clinical appointments to continuous, personalised, AI-guided daily management. For healthcare providers, this reduces clinical resource burden by enabling early nutritional interventions before complications arise. For mothers, it provides accessible, culturally relevant guidance in their own language. No existing product in the Sri Lankan market offers pre-consumption GDM risk scoring.

8.3 Competitive Advantage

Feature	Our System	MyFitnessPal	Generic GDM Apps	Manual Dietitian
Pre-consumption risk scoring	**Yes**	No	No	Manual
Sri Lankan food database	**Yes**	Partial	No	Yes
Sinhala voice input	**Yes**	No	No	N/A
Adaptive exercise guidance	**Yes**	No	No	Manual
Collaborative food learning	**Yes**	No	No	No
Real-time ecosystem integration	**Yes**	No	No	No
Gamification (SDT-based)	**Yes**	Basic	No	No

8.4 Revenue Model

- B2B Healthcare SaaS: Monthly subscription for clinic and hospital deployment — LKR 25,000–75,000 per facility depending on active user count.
- Direct-to-Consumer (Freemium): Core logging free; Premium tier (LKR 990/month) unlocks AI Risk Scoring, Virtual Chef, and doctor dashboard sharing.

- Health Insurance Partnerships: Data-sharing agreements with insurers who wish to incentivise GDM prevention in their member base.
- Commercialization of Collaborative Database: The accumulated Sri Lankan food nutritional database, once sufficiently large, can be licensed to other health applications.

9. BUDGET AND JUSTIFICATION

#	Item	Total (LKR)	Justification
1	Google Colab Pro (3 months)	13,500	YOLOv8 training on GPU; hyperparameter search
2	External hard drive (1TB)	3,500	Dataset backup: 500+ food images, model checkpoints
3	OpenWeatherMap API (Paid tier)	6,000	Reliable real-time temperature data for hydration logic
4	Test smartphones (rental)	5,000	Cross-platform Flutter testing on Android and iOS
5	Field data collection	8,000	Travel for food photography and participant recruitment
6	Participant compensation	10,000	Ethical compensation for user study participants (n=40)
7	Printing & documentation	3,500	Reports, posters, and final submission materials
	TOTAL	**LKR 49,500**	

All AI frameworks (Python, TensorFlow, Scikit-learn, FastAPI), the Flutter SDK, and Firebase free tier are open-source at zero cost. SLIIT laboratory computing provides sufficient power for model development outside training runs.

10. WORK BREAKDOWN STRUCTURE (WBS)

#	Task	Responsible	Duration
1	Literature review, dataset identification, Sri Lankan food database planning	Member 4	Weeks 1–3
2	Food image data collection (photography) and annotation	Member 4	Weeks 2–4
3	YOLOv8 model training and evaluation	Member 4	Weeks 4–6
4	OCR nutrition label pipeline + collaborative learning module	Member 4	Weeks 5–7
5	AI Risk Score engine + Deficiency detection module	Member 4	Weeks 6–8
6	Virtual Chef (Gemini API) + Hydration logic	Member 4	Weeks 7–9
7	Adaptive exercise logic + Pedometer/Accelerometer integration	Member 4	Weeks 8–10
8	Gamification system (points, badges, accountability alerts)	Member 4	Weeks 9–10
9	API integration with Member 1, 2, 3 data streams	Member 4 + Team	Weeks 10–11
10	Flutter frontend development + Sinhala voice integration	Member 4	Weeks 8–12
11	User study: compliance and accuracy testing (n=40)	Member 4 + Team	Weeks 12–13
12	Analysis, documentation, final report, and presentation	Member 4	Weeks 13–14

10.1 Key Milestones

Milestone	Target
Literature review complete; food database plan finalised	**Week 3**
YOLOv8 model trained with ≥ 85% accuracy on test set	**Week 6**
AI Risk Score engine validated against dietitian baseline	**Week 8**
Virtual Chef + Hydration + Exercise modules integrated	**Week 10**

Milestone	Target
Flutter app integrated with all co-researcher APIs	**Week 11**
User study complete; compliance data collected (n=40)	**Week 13**
Final report, poster, and demonstration ready	**Week 14**

11. GANTT CHART

The chart below uses filled (blue) cells to indicate active work periods for each task across the 14-week project timeline. A legend: filled cell = active work period.

#	Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
1	Literature Review & Dataset Collection	■	■	■											
2	Data Preprocessing & Feature Engineering			■	■										
3	Binary Classifier Development				■	■	■								
4	Multi-Source Data Fusion Module					■	■	■							
5	Priority Classification Model							■	■						
6	Online / Offline Mode Switching								■	■					
7	Confidence Scoring & REST API									■	■				
8	Integration with Server & Sensors										■	■			
9	Model Testing & Field Testing											■	■	■	
10	Analysis, Documentation & Report													■	■

■ *Blue cell = Active work period* □ *White/Grey cell = Inactive*

12. EVALUATION METRICS

Success will be demonstrated across three dimensions:

Dimension	Metric	Target
Technical Accuracy	YOLOv8 food recognition Confusion Matrix	Overall accuracy ≥ 90%; Precision ≥ 0.88; Recall ≥ 0.87
Technical Accuracy	OCR label extraction accuracy	≥ 95% match to ground-truth nutrient values
Technical Accuracy	Risk Score clinical agreement	≥ 85% triage agreement vs dietitian assessment
User Compliance	Logging frequency comparison (gamified vs control)	≥ 30% more meals logged per week in gamified group
User Compliance	System Usability Scale (SUS) score	SUS ≥ 80 (Good or above classification)
Clinical Impact	Post-prandial glucose correlation with AI-guided walks	Statistically significant reduction ($p < 0.05$) in glucose spike
Clinical Impact	Early deficiency flag lead time	Deficiency flagged ≥ 7 days before clinical presentation

13. REFERENCES

[1] Ministry of Health Sri Lanka, "National Nutrition Policy for Sri Lanka," Colombo, 2020.

[2] International Diabetes Federation, "IDF Diabetes Atlas — Gestational Diabetes," 10th ed., 2021.

[3] R. Retnakaran et al., "Association of Timing of Weight Gain in Pregnancy with Offspring Body Composition at 3 Years of Age," JAMA Pediatrics, 2019.

[4] B. Bhavadharini et al., "Gestational Diabetes Mellitus in South Asian Women," Obstetrics & Gynecology, vol. 135, no. 6, 2020.

[5] L. Mackillop et al., "Comparing the Efficacy of a Mobile Phone-Based Blood Glucose Management System in Gestational Diabetes," JMIR mHealth and uHealth, 2018.

[6] A. Meyers et al., "Im2Calories: Towards an Automated Mobile Vision Food Diary," Proc. ICCV, pp. 1233–1241, 2015.

[7] G. Jocher et al., "Ultralytics YOLOv8," GitHub Repository, 2023.

[8] G. Ciocca et al., "Food Recognition: A New Dataset, Experiments, and Results," IEEE Journal of Biomedical and Health Informatics, vol. 21, no. 3, pp. 588–598, 2017.

[9] D. J. A. Jenkins et al., "Glycemic Index: Overview of Implications in Health and Disease," American Journal of Clinical Nutrition, vol. 76, no. 1, pp. 266S–273S, 2002.

[10] T. M. S. Wolever, "Is Glycaemic Index (GI) a Valid Measure of Carbohydrate Quality?," European Journal of Clinical Nutrition, vol. 67, pp. 522–531, 2013.

[11] J. Hamari et al., "Does Gamification Work? A Literature Review of Empirical Studies on Gamification," Proc. HICSS, pp. 3025–3034, 2014.

[12] E. M. van Poppel et al., "Walking as an Intervention for Postprandial Blood Glucose in Gestational Diabetes," Diabetes Care, vol. 43, no. 2, 2020.

14. APPENDICES

Appendix A — Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
YOLOv8 accuracy insufficient for Sri Lankan dishes	Medium	High	Supplement Food-101 with 500+ manually photographed local dishes; iterative retraining with user-flagged misidentifications.
Participants drop out of user study	Medium	Medium	Recruit n=50 to buffer for 20% attrition; provide ethical compensation; keep study period to 2 weeks.
Gemini API rate limits or cost overrun	Low	Medium	Cache common recipe recommendations locally; implement request batching; set API spending cap.
Risk Score formula challenged by clinical reviewers	Medium	High	Frame as a decision-support heuristic, not a clinical diagnostic. Validate against dietitian assessments in Appendix B.
Integration API breaks with co-researcher systems	Medium	High	Define API contracts in Week 2; weekly integration checks from Week 9; mock API endpoints for independent testing.
Firebase security misconfiguration leaks health data	Low	Critical	Security rule audit by Week 6; penetration testing before user study; all data encrypted at rest.

Appendix B — Personnel and Facilities

Facility	Use
SLIIT IT Laboratory	AI model development, FastAPI backend, Flutter frontend, Firebase integration
SLIIT Electronics Lab	Sensor testing (accelerometer/pedometer integration with Flutter)
SLIIT Campus / Field	Food photography for dataset; user study sessions
Google Colab Pro (Cloud)	YOLOv8 model training and hyperparameter optimisation on GPU

Facility	Use
Team Members' Laptops	Development, testing, and documentation (zero additional cost)

— End of Proposal Document —

Member 4 — Multimodal Virtual Nutritionist | Department of Information Technology, SLIIT | March 2026